Bilingual Translation / 双语译文
Source / 来源: Salmi - 3 Mapping and Viewing History.pdf
Format / 格式: English original block followed by Chinese translation block.

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recent guidebooks have paved the way for this by spreading knowledge and knowhow on useful methodologies and the necessary computational diagrams behind them." As already pointed out in this chapter, the dominating role of textual sources in the digital humanities faces growing criticism, much as it does in digital history. Before having a closer look at visual and audio-visual materials in the next chapter, it is worth recalling that the supposed duality between textuality and visuality is quite misleading and should not be taken too strictly. Texts include images and vice versa. Today, Internet texts often include both sound and visuals. And, moreover, both texts and images can be considered as material remains of the past. Text is not an abstract order of characters but is always, to draw on Bruno Latour's idea, inscribed on material objects, on stone or marble, on paper or hard drives."' Texts are material entities that have dimensions: printed texts manifest themselves, for example, as folios and octavos, as broadsheets and tabloids.
When considering texts as big data, metatext must seriously be taken into account, since it offers access to the material dimensions of the texts at hand. Metadata has recently been successfully used in the study of book history, in which it is vital to comprehend how books circulated as concrete artefacts in the past.101 In the future, these material ramifications might be avenues for bringing the quantifiable data of texts more closely together with the study of their contents and, thus, their impacts on the imaginations of readers.
3 Mapping and Viewing History In the early stages of digital history, in the 1990s, maps were of particular interest, perhaps more so than the textual materials discussed in the previous chapter, because large-scale digitization projects had only recently begun. At the University of Virginia in 1993, William G. Thomas and Edward L. Ayers started a project on the US Civil War, entitled The Valley of the Shadow, that brought together the experiences of Confederate and Union soldiers. It published historical documents online, including letters, diaries, images, newspapers, church records, census records, tax records, statistics – and maps. The pre-war material, for example, included maps pertaining to geography, infrastructure, agriculture, politics, religion and sla

very in
the selected counties (Augusta in Virginia and Franklin in Pennsylvania).
These were not merely digitized pre-existing visualizations but also an integral
 part of
the project, produced for online use by the project members.' Visual sources hav
e been
at the heart of digital history from the start, and maps have been an essential
ingredient in visualizing archival data and establishing points for comparison.
The
following chapter continues by discussing maps further as sources and historical
interpretations, fuelled by the so-called spatial turn. The chapter proceeds by
selecting three other themes under discussion, the increased role of visual sour
ces,
including the idea of distant viewing,

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近年的指南书已经通过传播有用方法和必要计算流程方面的知识与操作经验，为这一领域铺平了
道路。正如上一章所指出的，文本性资料在数字人文学和数字史中的主导地位正在受到越来越多
的批评。在进一步考察视觉和视听材料之前，有必要提醒自己：文本性与视觉性之间所谓的二分
其实很容易误导人，不能过分严格地理解。文本可以包含图像，图像也可以包含文本；今天的互
联网文本还经常同时包含声音和视觉元素。文本和图像同样都可以被视为过去遗留下来的物质痕
迹。文本不是抽象的字符秩序，而总是铭写在石头、大理石、纸张或硬盘等物质载体之上。文本
也是有尺寸和形态的物质实体，例如印刷文本会以对开本、八开本、大开报纸或小报等形式出现
。把文本作为大数据来处理时，元文本也必须被认真纳入考虑，因为它提供了通往文本物质维度
的路径。元数据近来已被成功用于书籍史研究，帮助研究者理解书籍作为具体物品在过去如何流
通。未来，这些物质层面的影响也许能把文本的可量化数据与文本内容及其对读者想象力的影响
更紧密地联系起来。本章随后进入“地图与观看历史”。在20世纪90年代数字史早期阶段，
地图尤其受到关注，甚至可能比上一章讨论的文本材料更受重视，因为大规模数字化工程当时才
刚刚开始。1993年，弗吉尼亚大学的 William G. Thomas 和
Edward L. Ayers 开始了关于美国内战的 The Valley of
the Shadow 项目，把邦联与联邦士兵的经验汇集起来。该项目在线发布信件、日记
、图像、报纸、教会记录、人口普查记录、税务记录、统计资料以及地图。战前材料中包含有关
地理、基础设施、农业、政治、宗教和奴隶制的地图，覆盖弗吉尼亚的 Augusta
县和宾夕法尼亚的 Franklin 县。这些地图并不只是对既有可视化材料的数字化复制
，也是项目成员为网络使用而制作的核心组成部分。视觉资料从一开始就是数字史的核心，地图
则是可视化档案数据和建立比较点的重要工具。本章接下来讨论地图作为史料和历史解释，并以
所谓“空间转向”为背景，进一步讨论视觉资料作用的提升、远观、视听资料的多模态分析以及
元数据对于视觉史和视听史的可能性。

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recent developments in the multimodal analysis of audio-visual sources and the
possibilities of metadata for visual and audio-visual history.
Maps as digital sources Historians have always emphasized the meaning of space a
nd
spatiality. In the study of the past, where has been as important as when. This
involves, for example, locations
(such as cities and villages), natural landmarks (such as rivers, mountains or s
eas) and
both natural and political borderlines.
In addition to maps, spatiality has also been studied through an array of source
s, from
architectural plans and landscape images to verbal descriptions in travel accoun
ts and
fiction.
Historians have also theorized on the significance of space, as demonstrated in
Frederick Jackson Turner's famous essay
`The significance of the frontier in American history' (1893) and Fernand Braude
l's
classic The Mediterranean and the Mediterranean World in the Age of Philip II (1

949).
The so-called spatial turn in the humanities and social sciences had its roots in this enduring interest in spaces and places,' but in the 1990s it really entered the scholarly discourse. According to Doris Bachmann-Medick, this development was inspired by the end of the Cold War at the turn of the 1980s and 1990s; with the end of the political polarity of Eastern and Western blocs, a need arose to map the reorganization of the global order, to go beyond previous security strategies and to understand how transnational flows would change the way the world was imagined. Edward W. Soj''s Postmodern Geographies was a groundbreaking volume in 1989, followed by his Thirdspace in 1996. Soon, the spatial turn embraced several fields of study, including anthropology, archaeology, geography, gender studies and history.' For the digital historian, it is important to be aware of these backgrounds, but of course spatiality and its digital research methods are strongly linked to the development of geospatial information that has emerged at the same time as a result of multidisciplinary collaboration.
The digitization of historical sources described in chapter 1 has also included maps. Today, many national and international collections of digitized maps enable their study using both computational tools and more traditional methods.' For example, the Royal Library of Denmark had already begun scanning its map collection in 1997. At first, the idea was to make index maps to demonstrate the holdings and increase the usability of the collection. Another motivation was the challenge presented by oversized maps that were difficult to handle, and, of course, preservation concerns were also paramount because the use of digital copies would allow the originals to remain untouched. Since 2003, the library has offered rare maps through the Internet.' The National Library of Argentina now has 1,122 downloadable historical maps on its website, and the digital collection features not only older maps but also a substantial amount of material from the twentieth century.' One of the largest current collections of historical maps, plans and views, comprising 4.5 million items, is located at the British Library.
It includes not only scanned maps but also geospatial data, cartographic applications and digital aerial photographs, and it features both digitized and born-digital content.' The chronological range of the collection is over 2,000 years, and its size is suggested by the example of the King George III Topographical and Maritime collection, which alone includes `30-40,000 maps, plans and views, both printed and hand-drawn, of all parts of the world'.9 Not all maps are two-dimensional, however, and the British Library also holds hundreds of terrestrial and celestial globes, which are three-

dimensional material artefacts.10 As sources, maps are layered cultural artefacts that include numbers, texts, symbols and images. Historical research draws on these sources, which can be objects of careful close reading and iconographic analysis or can be used for more distant reading via the computational study of visual concepts or textual content. Maps can be described as representations of the physical world, but they also construct the world as we know it. Historical maps must always be understood in the context of the practices that originally contributed to their production, including the ways of measuring and representing distances and the strategies for aggregating data for maps; of course, the motives and uses of maps also influence the interpretation of these documents.

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地图作为数字史料。历史学家一向重视空间和空间性；在研究过去时，“何地”常常与“何时”同样重要。这包括城市和村庄等地点，河流、山脉、海洋等自然地标，以及自然边界和政治边界。除了地图之外，空间性还可以通过多种资料研究，如建筑图、风景图像、游记和小说中的文字描述。历史学家也对空间的重要性进行过理论化阐释，例如 Frederick Jackson Turner 的《美国历史中边疆的重要性》和 Braudel 的《菲利普二世时代的地中海和地中海世界》。人文与社会科学中的“空间转向”根植于这种对空间和地点的长期兴趣，但它在20世纪90年代真正进入学术话语。Doris Bachmann-Medick 认为，这一发展受到冷战结束的启发：东西方集团的政治两极格局瓦解后，人们需要重新描绘全球秩序，超越旧的安全战略，并理解跨国流动如何改变世界想象。Edward W. Soja 的 Postmodern Geographies 和 Thirdspace 是重要著作。空间转向很快影响了人类学、考古学、地理学、性别研究和历史学等领域。对数字历史学家来说，了解这些背景很重要，同时空间性及其数字研究方法也与地理空间信息的发展紧密相连，而这种发展本身就是跨学科合作的结果。第一章所说的历史资料数字化也包括地图。今天，许多国家和国际地图馆藏都已数字化，研究者可以用计算工具和传统方法研究它们。丹麦皇家图书馆早在1997年就开始扫描地图；起初是为了制作索引图，展示馆藏并提高可用性，同时也出于处理超大地图和保护原件的需要。阿根廷国家图书馆提供了可下载的历史地图，英国图书馆拥有规模巨大的历史地图、平面图和景观图收藏，并包括扫描地图、地理空间数据、制图应用、数字航拍照片以及数字化和数字原生内容。地图作为史料，是包含数字、文本、符号和图像的多层文化物。它们既可以被细读和图像志分析，也可以通过计算方法对视觉概念或文字内容进行较远距离的研究。地图可以说是物理世界的再现，但同时也建构了我们所认识的世界。因此，历史地图必须放回产生它们的实践语境中理解，包括测量和表现距离的方式、把数据聚合成地图的策略，以及地图的动机和用途。

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Maps possess a strong representational character as they usually try to capture the three-dimensional world in only two dimensions. In Exploring Big Historical Data, Shawn Graham, Ian Milligan and Scott Weingart point out that `these seemingly straightforward representations are loaded with difficult choices, as laying two-dimensional coordinates onto a 3D world means making complicated choices around what map projection to use'." They note, for example, that, in the Mercator projection that underlies Google Maps, Greenland looks almost as big as Africa, although Africa is, in reality, fourteen times larger.12 Present-day projections differ i

n how
they address the problem of 2D/3D conversion, and many other factors influenced how old
maps projected the physical world. In the past, maps also represented a realm of
imagination, and their environmental features, such as coastlines and riverbeds, may
have transformed over time, so it is not always possible to reduce historical maps to
the topographical world. The researcher must estimate the possibility of matching a
historical map with a modern one and determine whether this serves the purpose of the
scholarly task. If so, the next step is trying to identify the points on the historical
map that correspond to latitudes and longitudes in the present-day reference map (a
process called georectification). Since the 1980s, geographical information systems
(GIS) have provided tools for managing geographical information. GIS makes it possible
to draw on historical maps by extracting their information and analysing them in
comparison with other GIS data and data sets.13 The New York Public Library's collection
of over 11,000 georectified historical maps is openly accessible to users.14 Maps as
historical interpretations The use of historical maps (and in fact any source) depends
on the research problem. Sometimes, georectification is not an option, for example in
the case when no actual historical map is available. Other possible sources of
topographical and geographical information can be further processed into a digital map,
presenting a historical interpretation.
The historian Harri Kiiskinen drew on GIS in his study of the production and trade of
terra sigillata pottery in Roman Etruria. To understand the trading routes, he tried to
determine which rivers in Etruria were navigable at the time,
2,000 years ago. In addition to archaeological and historical evidence, this work used a
digital terrain model based on topographic data from NASA's Shuttle Radar Topography
Mission. As a result, Kiiskinen was able to draw a map that reconstructed the river
network in Etruria.15 Here, the digital map serves research purposes and expresses an
important historical argument.
Maps had already been used to visualize big historical data before the era of GIS and
the digital humanities. An excellent reminder of this is the production of Charles-
Joseph Minard
(1781-1870), a French civil engineer who won fame for his informational graphics and
maps.16 His most well-known map, from 1869, describes Napoleon's Russian campaign of
1812. There is a geographic orientation in the image of the direction of the campaign,
but the map incorporates many other elements of information, from the number of troops
to the temperature." The map clearly shows how Napoleon's army suffered during its march

towards' Moscow from the Russian border. Minard also captured phenomena of his own era,
for example, migratory movements on a global scale.
The nineteenth century was similar to today in that it was characterized by a sudden
abundance of data provided by new technologies, such as the high-speed press, the
railroad and the telegram." Today, Minard is often recognized as the developer of the
so-called flow map; in 1845, spurred by the discussion of the development of railroad
connections, he produced a map that showed how passengers circulated in the road routes
between Dijon and Mulhouse.19 Nineteenth-century statistics have had a remarkable
influence on how we visualize information. Graham, Milligan and Weingart mention the dot
density map that was used, for example, by the English physician John Snow (1813-1858),
who, drawing on statistical methods, revealed the correlation between contaminated water
and cholera. Snow marked the locations of London cholera cases with a dot on a map and
was able to show that many cases occurred around a water pump.2°

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地图有很强的再现性，因为它通常试图用二维形式捕捉三维世界。Graham、Milligan 和 Weingart 指出，看似直观的地图再现其实充满困难选择，因为把三维世界投射为二维坐标必然涉及地图投影的复杂取舍。例如 Google Maps 所基于的墨卡托投影会让格陵兰看起来几乎与非洲一样大，尽管非洲实际面积约为格陵兰的14倍。今天的投影方式以不同办法处理二维和三维转换，旧地图呈现物理世界的方式也受多种因素影响。过去的地图还包含想象领域，海岸线和河床等环境特征也可能随时间变化，因此并不总能把历史地图简单还原为地形世界。研究者必须判断历史地图能否与现代地图匹配，以及这种匹配是否服务于研究任务。如果可以，下一步就是识别历史地图上与现今参考地图经纬度对应的点，这一过程称为地理校正。自20世纪80年代以来，地理信息系统（GIS）为地理信息管理提供了工具，使研究者能够提取历史地图信息，并与其他 GIS 数据和数据集比较分析。纽约公共图书馆开放了超过11000幅经过地理校正的历史地图。地图也可以作为历史解释来使用。是否采用历史地图取决于研究问题；有时没有真实历史地图可供校正，研究者就需要把其他地形和地理信息加工成数字地图，从而呈现一种历史解释。Harri Kiiskinen 在研究罗马伊特鲁里亚陶器 terra sigillata 的生产与贸易时使用 GIS，为理解贸易路线，他结合考古、历史证据和 NASA Shuttle Radar Topography Mission 的地形数据，重建了两千年前伊特鲁里亚可通航河网。这里的数字地图既服务研究，也表达重要的历史论点。在 GIS 和数字人文学出现之前，地图已被用于可视化历史大数据。Charles-Joseph Minard 1869年关于拿破仑1812年俄国战役的名图不仅呈现行军方向，还把兵力和温度等信息合并其中，清楚显示军队在行进中的损耗。19世纪与今天相似，也经历了由高速印刷、铁路和电报等新技术带来的数据骤增。John Snow 关于伦敦霍乱病例与水泵关系的点密度图也深刻影响了信息可视化。

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Although recognizing these roots is important, digital technologies obviously offer many
new avenues for maps, mapping and data visualization in general. Digital maps enable
zooming in and out so that they can function as platforms for accumulated data for
various historical sources.
For example, a city map can include textual information and images of historical places

for the user. The map can also be a channel for simultaneously presenting diverse research results. Minard did the same in his infographics, but digital maps are not as entangled with the two-dimensional world.

Virtual maps can represent a third dimension and thus provide a broader visual perspective on the past.

From visual evidence to distant viewing Thus far, this chapter has dealt with maps and mapping, which have been at the heart of digital history since the 1990s. What must be discussed next is actual visual evidence from the past, including how it can contribute to and change our understanding of history. Today, digital historians have access to manuscript illustrations, photos, posters and works of art, but digitization rates vary in collections that sometimes offer free use and sometimes only restricted use. For many reasons, the boundary between free and restricted use also varies. For example, there may be copyright concerns that influence the availability of images. In museum collections, copyright holders are not only artists but also photographers who have produced digital reproductions.

One of the most well-known examples of making visual content available is the Rijksmuseum in Amsterdam, which is a Dutch national museum that holds a collection of over a million objects, including 2,000 paintings. In 2012, the Rijksmuseum started an extraordinary project by making 125,000 high-resolution images available and downloadable. The plan is to add 40,000 images annually until the entire collection is online. Users can explore these images through the museum's own Rijksstudio, make observations, take clips and use them both in research and for other purposes. At the time of writing (May 2020), there were as many as 676,693 high-resolution images available through Rijksstudio.21 The collection is a substantial window, not only onto the Dutch past but also onto art history in general.

Restricted collections can also be useful for research because their metadata are often available online. An excellent example is provided by RMN Photo. RMN refers to the French organization Reunion des musees nationaux, a joint enterprise of French art galleries and museums. Today, RMN Photo also includes foreign galleries, such as the Prado in Madrid and the National Galleries of Scotland in Edinburgh.22 The database includes small preview images that allow users to peek at the collection, and it is also possible to search by keyword. The collection is immense; a search of the keyword arbre (tree), returns over 22,000 images of various kinds of trees, including paintings, drawings, photographs, statues and many other kinds of objects from the Middle Ages to the present. This includes, for example, over 1,400 paintings from the nineteenth century.

In addition to RMN Photo and Amsterdam's Rijksmuseum, numerous national and

international image collections can be put under the lens. National libraries around the world have digitized cultural artefacts and images, and many heritage projects offer visual content to users, from the American Memory project to Europeana. The latter, for example, contains transnational collections, such as First World War newsreels, postcards and letters. The film collection alone comprises as many as 2,726 items that can be regarded as one of the most comprehensive cinematic collections on the Great War.23 It is obvious that these collections set different premises and conditions for research than textual material, but they also offer many possibilities for digital historians.

In his research lab Software Studies Initiative, Lev Manovich and his team have processed massive collections of images and video, including 6,000 paintings by French Impressionists and 20,000 photographs from the photography collection of the Museum of Modern Art. From a historical perspective, the Impressionist collection of images is comprehensive, as it represents approximately 50 per cent of all paintings and pastels presented in the Impressionist exhibitions during the years 1874-1886. The experiment of the Software Studies Initiative started with the idea of extracting particular features – in fact, 200 of them from

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认识这些历史根源固然重要，但数字技术显然为地图、制图和数据可视化开辟了新路径。数字地图可以缩放，因而可以作为汇聚多种历史资料数据的平台。城市地图可以为用户包含历史地点的文字信息和图像，也可以同时呈现多样研究结果。Minard 在信息图中做过类似尝试，但数字地图不再那么受二维世界限制；虚拟地图可以表现第三维度，从而提供更广阔的历史视觉视角。接下来，本章从地图转向来自过去的实际视觉证据，讨论它们如何帮助并改变我们对历史的理解。今天，数字历史学家可以接触手稿插图、照片、海报和艺术作品，但不同馆藏的数字化程度和使用权限差异很大。版权问题会影响图像可用性；在博物馆收藏中，版权持有人不仅包括艺术家，也包括制作数字复制品的摄影师。阿姆斯特丹国立博物馆是开放视觉内容的著名案例。2012年起，该馆开放并允许下载大量高分辨率图像，并计划持续增加。用户可以通过 Rijksstudio 浏览、观察、截取并在研究或其他目的上使用这些图像。到2020年5月，Rijksstudio 已有676693张高分辨率图像，这一收藏不仅为荷兰过去，也为艺术史整体打开了重要窗口。受限馆藏同样有研究价值，因为它们的元数据常常在线可用。RMN Photo 是一个例子，它隶属于法国国家博物馆联盟，后来也收入普拉多美术馆、苏格兰国家美术馆等外国机构。该数据库有小型预览图，并可按关键词搜索；仅 arbre（树）一词就返回超过22000幅关于树的图像，涵盖绘画、素描、照片、雕塑等多种对象。除 RMN Photo 和 Rijksmuseum 外，世界各地国家图书馆和文化遗产项目也提供大量图像，从 American Memory 到 Europeana。Europeana 的一战新闻片、明信片和信件等跨国收藏为研究提供了不同于文本材料的前提和条件，也给数字历史学家带来丰富可能。Lev Manovich 的 Software Studies Initiative 处理过大量图像和视频，包括法国印象派绘画和现代艺术博物馆摄影收藏，并尝试从数据集中提取大量特征。

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the data set. This included `colour characteristics, contrast, shapes, textures and some aspects of composition'.24 These features were then computationally reduced to a

smaller
number of dimensions to enable the grouping of images and thus the viewing of them from
a distance, as a whole. As a result, they found that the types of images that are
usually associated with Impressionism, with lighter tones, were a smaller part of the
data set. As Manovich writes, `At least half of the images turn out to be rather
traditional and more typical of classical 19th century painting (darker tones and warm
colors).'' Manovich presented this experiment in his article `Data science and digital
art history', published in 2015. The actual aim of the text was to articulate basic
concepts for the study of large image corpora, drawing on data science and especially
considering what the extraction features, such as average brightness and saturation,
could offer to the digital analysis of large data sets of images. In 2019, Taylor Arnold
and Lauren Tilton emphasized strongly what they call distant viewing:
[W]e argue that DH [digital humanities] should consider, what we call, distant viewing –
a methodological and theoretical framework for studying large collections of visual
material. Distant viewing is distinguished from other approaches by making explicit the
interpretive nature of extracting semantic metadata from images.
In other words, one must `view' visual materials before studying them. Viewing, which we
define as an interpretive action taken by either a person or a model, is necessitated by
the way in which information is transferred in visual materials." Manovich did not use
the expression `distant viewing', but this was exactly the aim of his texts: to try to
track down what machine viewing could mean and how it could be used in the digital
humanities. In recent years, `distant viewing' has been employed by other researchers as
well,27 which shows the increasing interest in the computational study and cultural
analytics of images. In the case of distant reading, the question is: what does it mean
when a machine `reads' a text, which features have to be computationally extracted and
how could these features be used for analysis? The same goes for the analysis of images,
both still and moving: what does
`viewing' mean? This is what Arnold and Tilton underline in their article: there is a
`need for coding visual materials prior to exploratory data analysis of visual
corpora'.' At the same time, it is important to recognize that there are different
genres and different kinds of images; therefore, there are particular characteristics
that the interpretation should take into account.29 Sound and vision Parallel to the
previous discussion, in data science, and not always in connection to the digital
humanities, much research has taken place in relation to sound and vision. Automatic

speech recognition (ASR) is today a useful tool for the study of audio-visual material.
It enables the extraction of speech and its convergence into text, which again can be
analysed with text mining tools. Simultaneously ASR processing can be deployed for
producing digital indexes of the contents of auditive or audio-visual sources and thus
help to enhance existing metadata.
Considering how much audio material is in the archives, computational methods could be
employed to increase their accessibility and, of course, to study en masse sources that
would otherwise be difficult to digest. As a research example, we could play with the
idea of applying ASR to newsreels after the breakthrough of sound cinema in the late
1920s and early 1930s. Often, these newsreels were made with a silent camera, without
any sound recorded on location. A voiceover narration was added later in the studio.
Newsreels were a text-driven form of audio-visual culture; therefore, speech extraction
might offer illuminating results. In any case, the quality of sound, and possible
background music, would make this research challenging. Another problem might be the
fact that ASR draws heavily on language models in identifying words. It seems that there
is insufficient research on historical language models, which from a historian's

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这些特征包括色彩特征、对比度、形状、纹理和构图的某些方面，随后通过计算方法降维，以便把图像作为整体进行分组和远距离观看。结果显示，通常与印象派联系在一起的浅色调图像其实只占数据集较小部分；至少一半图像更传统，更接近19世纪古典绘画的暗色和暖色。Manovich 在2015年《数据科学与数字艺术史》中介绍了这一实验，目的在于为大型图像语料研究提出基本概念，尤其思考平均亮度、饱和度等特征提取能为大型图像数据集分析提供什么。2019年，Taylor Arnold 和 Lauren Tilton 强调了“远观”这一概念：他们认为数字人文学应考虑一种研究大型视觉材料集合的方法论和理论框架，即远观。远观与其他方法的区别在于，它明确承认从图像中提取语义元数据本身具有解释性。换言之，在研究视觉材料之前，必须先“观看”它们；这种观看可以由人或模型完成，并且由视觉材料传递信息的方式所要求。Manovich 虽未使用“远观”一词，但他的文本正是在追问机器观看意味着什么以及如何用于数字人文学。近年来，“远观”也被其他研究者采用，表明人们越来越关注图像的计算研究和文化分析。与远读相似，关键问题是：机器“阅读”文本时意味着什么，哪些特征必须被计算提取，这些特征又如何用于分析？对静态和动态图像亦然：“观看”究竟意味着什么？Arnold 和 Tilton 强调，在探索性分析视觉语料之前，需要先对视觉材料编码。同时，也必须承认不同图像类型和体裁具有不同特征，解释时应予以考虑。关于声音与视觉，数据科学中已有大量研究，未必总与数字人文学直接相连。自动语音识别（ASR）如今是研究视听材料的有用工具，它可以把语音提取并转换为文本，再用文本挖掘工具分析；ASR 也可以用来为音频或视听来源制作数字索引，增强现有元数据。档案中有大量音频材料，计算方法可以提高其可访问性，并使研究者能够批量分析本来难以消化的资料。例如可以设想把 ASR 用于20世纪20年代末和30年代初有声电影突破后的新闻短片。许多新闻短片现场以无声摄影机拍摄，旁白后来在录音棚加入，因此它们是一种文本驱动的视听文化，语音提取可能带来启发。不过音质、背景音乐和语言模型都会带来挑战，历史语言模型研究仍显不足。

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perspective would be essential for recognizing speech acts of the past. Not only
language but also the way we speak, our enunciation, has changed tremendously. I

f we
think of fictitious material, such as narrative films from the 1930s and
1940s, the dialogue might also include different dialects and slang. Furthermore, the
quality of sound technology and the sounds of the filming location would make the
recognition even more painstaking. Still, the idea to be able to mine and to distant-
view old cinematic content is tempting, and it would offer an avenue into the
imagination of past decades.
There are sophisticated technologies that can further help to refine methodologies, such
as audio event detection, voice activity detection and speaker diarization, which can
help to identify who actually speaks and even connect the voice to the speaker when he
or she is not visible on screen.30 Another area of recent exploration is the visual
analysis of movie content, drawing on the technology around deep neural networks
(DNNs).
This includes, for example, such research strategies as visual feature and facial
expression extraction and visual content description, which go beyond visuality into the
multimodal analysis of audio-visual sources.31 The development of content detectors
helps to identify details in big audio-visual data sets. If the research project spans a
long period, the analysis of movie content would give raw material, such as on how the
environments, milieus and cultural artefacts used in these works change over time.
In the study of audio-visual content, it is necessary to pay attention to the changing
technologies after the nineteenth century: they influence the ways in which these
cultural products can be used for digital analysis and how they can be disseminated.
Early film history was preserved on nitrate cellulose material which was very fragile
and flammable.
Therefore, many of the early films have been lost, and their study must be based on
other kinds of sources, including stills, manuscripts, newspaper ads and reviews.
Acetate cellulose is more stable and became dominant after the Second World War. Film
archives around the world have for decades aimed at preserving these copies, first by
producing acetate duplicates, now by making digital scans with 4k or even higher
resolution. After the advent of video tape recorders in the
1950s, many other technologies for audio-visual products have been introduced. In the
1980s Betamax and VHS technology dominated consumer markets, but there were also other
formats, such as Video 2000 and U-matic. In the 1990s, digital CDs, laserdiscs and DVDs
came along, followed by Blu-ray discs and streaming services in the 2000s. In this time
period resolutions also changed, from high definition
(HD) to ultra-high definition (UHD) and from 2k to 4k or even 8k. The array of
historical technologies is a challenge for the digital historian, interested in

audio-
visual content.
Film archives have produced digital copies from these items but there are many p
roblems
to be solved, for example, in the use of home-made movies that can still be pres
erved in
their original format.
Visuality, textuality and metadata As a conclusion to this chapter, I would like
 to
return to two other aspects in the study of visuality that should be remembered
and
remind us of the closeness to the study of textuality. The first issue is that t
he
dominant part of digitization projects mentioned in this book is in fact based o
n
images. Digitized books, journals and newspapers originate from image collection
s, as
printed material too is first scanned into an image and only thereafter is the t
ext
recognized using OCR. This means, for instance, that all newspaper repositories
are
basically collections of images of newspapers, although this is only seldom high
lighted,
as most research methods emphasize texts to which OCR has been applied. Yet, the
re are
many other opportunities in these resources. Textuality is not the opposite of
visuality, but texts are visual entities in themselves and comprise colour, font
 type,
font size, spacing and many other layout elements. The historical change of a fr
ont
page, to name only one possible theme, could be explored based on the scanned ma
terial.
How did headlines change over time? How many columns were there? When did drawin
gs and
photographs first appear on the cover? How is the role of advertisements transfo
rmed if
analysed in a historical setting? In addition to actual scanned images, visualit
y can be
approached through metadata. Usually, XML files include information on the

从历史学角度看，识别过去的言语行为需要历史语言模型；不仅语言变了，人们的发音和说话方
式也发生了巨大变化。如果研究20世纪30年代和40年代的叙事电影，对话还可能包含不同
方言和俚语；录音技术质量和拍摄现场声音也会让识别更困难。尽管如此，挖掘并远观旧电影内
容仍很有吸引力，因为它能通向过去几十年的想象世界。音频事件检测、语音活动检测和说话人
分离等技术可以进一步改进方法，帮助识别谁在说话，甚至在说话者不出现在画面中时把声音与
人物连接起来。另一个新近探索领域是借助深度神经网络（DNN）进行电影内容的视觉分析，
包括视觉特征提取、面部表情提取和视觉内容描述等，这些方法超出单纯视觉性，进入视听资料
的多模态分析。内容检测器的发展有助于在大型视听数据集中识别细节；若研究跨越较长时期，
电影内容分析可以提供环境、场景和文化物品如何随时间变化的原始材料。研究视听内容时，还
必须注意19世纪以来不断变化的技术，因为它们影响这些文化产品如何被数字分析和传播。早
期电影保存在易碎、易燃的硝酸纤维素材料上，因此许多早期影片已经遗失，研究只能依赖剧照
、手稿、报纸广告和评论等其他资料。更稳定的醋酸纤维素在二战后占主导。世界各地电影档案
馆长期致力于保存这些复制品，先制作醋酸复制片，现在则以4K或更高分辨率进行数字扫描。
20世纪50年代录像机出现后，视听产品技术持续更迭：80年代 Betamax 和
VHS 主导消费市场，也有 Video 2000 和
U-matic；90年代出现数字 CD、激光影碟和
DVD；2000年代则有蓝光和流媒体。分辨率也从 HD 到 UHD，从2K到4K甚至
8K。历史技术的多样性对研究视听内容的数字历史学家构成挑战。电影档案馆虽已制作数字副
本，但仍有许多问题待解决，例如如何使用仍以原始格式保存的家庭电影。章节最后回到视觉性
、文本性与元数据。许多数字化项目实际上以图像为基础：书籍、期刊和报纸先被扫描成图像，

再通过 OCR 识别文字。因此，报纸数据库本质上是报纸图像集合，只是多数方法强调
OCR 后的文本。文本性并非视觉性的反面，文本本身就是视觉实体，包含颜色、字体、字号
、间距和版面等元素。研究者可以利用扫描材料研究头版如何变化、标题如何变化、栏数如何变
化、图画与照片何时出现在封面，以及广告角色如何在历史中转变。除了扫描图像，视觉性也可
通过元数据来研究；XML
文件通常包含文本和图像在文档中的视觉内容与物理位置，可帮助定位报刊中的图画和照片。

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visual contents and physical places of the texts and images on the document. This data
would help the researcher to locate the drawings and photos that have been published in
newspapers and magazines.
As the last aspect of this chapter, a discussion on metadata is needed, but in another
sense. All databases of images contain metatextual information on each object. There is
certainly variation in the quality of this data, but it will be informative in multiple
ways. Let us take the example of the online databases of art museums. Tretyakov Gallery
in Moscow has one of the most famous collections in the world, including over 180,000
works of art. The online collection includes the name of the artist and painting, the
measurements of the artwork and the technique used for the work.' These are standard
details, but they give the option for the researcher to explore the proportions of the
collection, for example, how particular techniques change over time. Some databases
consist of many more details. Let me continue with an example presented already earlier
in this chapter. One of the most comprehensive databases on European art is RMN Photo,
which began in 1946 as a commercial institution under the authority of the French
Ministry of Culture. Its original idea was to promote the collections of the national
museums of France, and it is continuing this mission; although it offers access to some
foreign collections as well. Today, it comprises almost 800,000 photos of artworks
housed in national and regional museums, including the Louvre, the Musee d'Orsay, the
Centre Pompidou, Chantilly's Conde Museum and Palais de Beaux Arts de Lille. All these
images are accessible online, although high-resolution copies can only be purchased.33
The metadata of each item has been enriched with many details, including keywords. For
example, the painting Pauline, femme de Seneque, rappelie a la vie by Jean-Joseph
Taillasson
(1745-1809) is accompanied with an identifier (91-002122), inventory number (INV 8081),
period (eighteenth century, modern age), technique and material (oil on canvas), location
(Paris, Musee du Louvre), credit for the photo (RMN-Grand Palais, Musee du
Louvre/Christian Jean), size of the digital image (in this case 10535 x 8192 pix

els) and
keywords that relate to the content of the painting. This artwork portrays a fai nting
figure who is obviously the protagonist of the picture, Pauline. With the keywor d
`fainting', the user can detect other images of fainting. The database shows tha t RMN
Photo covers seventeen other fainting images from the eighteenth century. In add ition,
the collection includes 178 artworks that portray fainting, most of them oil pai ntings,
but there are also photographs, textiles, pencil drawings, ink works and so on. This
example in itself is only one case for close reading, but if this whole database could
be analysed, it would provide an exceptional view into the development of art, a s well
as allow other kinds of more specific research settings, such as the history of fainting.
Visual and audio-visual sources can be productively approached through metadata. This
requires access to the database as a data dump. One of the most comprehensive
international filmographies available is the Internet Movie Database (IMDb). It has been
criticized for its incompleteness and inaccuracies, but its great benefit is its truly
transnational and transcontinental coverage, not only of films made in the past but also
new productions that are under preparation.
IMDb was launched in 1990 as a fan-based enterprise, but today it is a subsidiar y of
Amazon and delivers online massive amounts of content on films, television progr ammes
and video games. By January 2020, it included 6.5 million titles, including epis odes of
television series, and 10.4 million personality entries. For researchers, it is valuable
that this data is largely made available through open access. IMDb offers seven
different data sets that are refreshed on a daily basis.34 These data have 'alre ady been
used for historically oriented studies. For example, Bo McCready has visualized the
popularity of film genres between 1910 and 2018 and described the development of twelve
genres, from action films to western movies.35 This is obviously only a scratch on the
surface of what could be done and achieved. The database could be employed in an alysing
the networks of actors, directors and other production staff; movements of peopl e by
connecting movie titles with production countries;
changes in production volumes on a transregional and global level; and so on. Fo r a film
scholar, this would also offer the possibility to glimpse into the future, as th e
database also covers on-going productions. This sketch on the resources of IMDb is of
course only one option out of many for writing

最后，本章从另一层意义讨论元数据。所有图像数据库都包含每个对象的元文本信息，虽然数据

质量不同，但可以在多方面提供信息。以美术馆在线数据库为例，莫斯科特列季亚科夫画廊拥有世界著名收藏，在线馆藏包含艺术家和作品名称、作品尺寸及技法等标准信息，研究者由此可考察馆藏比例和特定技法随时间的变化。一些数据库提供更丰富细节。RMN Photo 是欧洲艺术的重要数据库之一，1946年作为法国文化部管理下的商业机构开始，原本旨在推广法国国家博物馆收藏，如今也提供部分外国收藏。它包含卢浮宫、奥赛博物馆、蓬皮杜中心、尚蒂伊孔代博物馆和里尔美术宫等国家与地区博物馆近80万张艺术品照片。图像均可在线访问，高分辨率副本则需购买。每件作品的元数据都经过丰富，包括关键词。Jean-Joseph Taillasson 的 Pauline, femme de Seneque, rappelie a la vie 这幅画就附有标识符、库存号、时期、技法与材料、所在地、照片署名、数字图像尺寸和内容关键词。作品描绘了昏厥人物Pauline；用户用“fainting”检索即可发现其他昏厥图像。RMN Photo 中还有17幅18世纪昏厥图像，整个收藏则有178件描绘昏厥的艺术品，包括油画、照片、纺织品、铅笔画、墨水作品等。单件作品只是细读案例，但若能分析整个数据库，就能以异常丰富的方式观察艺术发展，也能提出更具体问题，如昏厥史。视觉和视听资料都可以通过元数据富有成效地研究，但这要求能够以数据包形式访问数据库。IMDb 是最全面的国际影视资料库之一，虽因不完整和错误受到批评，却具有真正跨国、跨洲覆盖的优势，不仅包括过去的影片，也包括正在筹备的新作品。IMDb 1990年以影迷项目起步，如今隶属于Amazon，提供大量电影、电视节目和电子游戏信息。到2020年1月，它有650万条标题记录和1040万条人物记录，并以开放访问方式提供七个每日更新的数据集。研究者已用这些数据进行历史取向研究，例如 Bo McCready 可视化了1910至2018年电影类型的流行变化。未来还可用于分析演员、导演和制作人员网络，连接影片和生产国家来研究人员流动，以及从跨地区和全球层面研究生产量变化。

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an audio-visual history. It could later be joined with other databases that include more
information on national cinemas and television broadcasts. Finally, the content itself,
actual films and television programmes are increasingly available in digital format.'
Digital technologies of visualization have been, and continue to be, an expanding area
of research and development in the
2000s. Visual devices penetrate our present world, from digital cameras and mobile
phones to UHD television sets and high-resolution displays. The use of these
technologies has taken a central position in how we entertain ourselves, in how we
express our relationship with the world around us, and how we are controlled and
policed. During recent decades, audio-visual culture has undergone a transformation –
from analogue movie theatres to 4k digital cinemas
- to such an extent that nitrate and cellulose acetate films need special arrangements
and old technology to be able to be shown. Consumers today produce visual and audio-
visual content more than ever before, and they share these visualizations on social
media and preserve many more on their private hard drives, as well as on cloud services.
To be sure, the years to come will bring forth new insights, methods and applications
that interpret history both in visual terms and based on visual cultural heritage.
4 Interdisciplinarity:
Challenges for Research Vast resources are available for the digital historian. As the
previous discussions show, at the same time as more material is easily available, there
are many heuristic and source-critical aspects that have to be taken into consid

eration.
A major challenge is the use of large data sets. Therefore, digital history ofte
n
appears as interdisciplinary work and takes the form of collaborative research b
ased on
cooperation and joint work. There is a need for transgressing disciplinary borde
rs not
only because of the complex problems awaiting resolution but also because, in ot
her
fields of research, such generic tools have been, or might have been, developed
and
productively deployed that can be applied in other contexts. Digital historians
think
about their work in terms of their interest in historical problems, but already
problem-
solving might require expertise from different disciplines and methodological
traditions. It would be unfortunate not to take advantage of what has been found
in
other scientific communities and within different research paradigms.
Digital history does not stand alone before this challenge.
The world has always been complex, to be sure, but we are increasingly conscious
of the
global hazards we confront, which further emphasizes the need for interdisciplin
ary
work.
In his compelling book We Have Never Been Modern,

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IMDb 只是书写视听史的众多资源之一，未来还可与包含更多国家电影和电视广播信息的数据库连接。内容本身，即实际电影和电视节目，也越来越多地以数字形式存在。数字可视化技术在21世纪一直是不断扩展的研究与开发领域。视觉设备深入当下世界，从数码相机、手机到UHD 电视和高分辨率显示器；这些技术在娱乐、表达我们与世界关系以及被控制和警务化方面都占据核心位置。近几十年来，视听文化已经从模拟电影院转向4K数字影院，以至于硝酸片和醋酸纤维素影片需要特殊安排和旧技术才能放映。今天消费者比以往任何时候都更多地生产视觉与视听内容，把它们分享到社交媒体，并把更多内容保存在私人硬盘和云服务上。未来必将出现新的洞见、方法和应用，以视觉方式并基于视觉文化遗产来解释历史。文件末尾进入下一章“跨学科性：研究的挑战”。数字历史学家拥有庞大资源，但材料更容易获取的同时，也必须考虑许多启发式和来源批判问题。大型数据集的使用是一项主要挑战，因此数字史常表现为跨学科工作，并采取协作研究形式。跨越学科边界不仅因为待解决问题复杂，也因为其他领域已经或可能已经开发出可迁移的通用工具。数字历史学家当然以历史问题为核心思考工作，但解决问题往往需要不同学科和方法传统的专业知识。忽视其他科学共同体和研究范式中的成果将是遗憾的。数字史并不是唯一面对这一挑战的领域；世界一直复杂，而我们也越来越意识到全球风险，这进一步凸显跨学科工作的必要性。